



Some thesis title

by

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Someone.

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* Supervisor X.

* Funding Y.

“ *A very long quote*
from someone very important. ”

– Mr M

Thank you to Leanne for helping to prepare this template.

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ABSTRACT

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NOMENCLATURE

ACRONYMS AND ABBREVIATIONS

MFT My Funky Thesis

DEFINITIONS

S

STUDENT is an entity needing a thesis to transcend the state of being a student.

INTRODUCTION

1.1 PROBLEM STATEMENT

This project will determine whether using Dynamic Time Warping (DTW) in the frequency domain, thus using vector features, is able to effectively detect Bryde's Whale calls within the False Bay region. Three feature extraction methods will be compared. These will be Short Time Fourier Transform (STFT), Mel-frequency Cepstral Coefficients (MFCC) and Linear Predictive Coding (LPC).

1.2 OBJECTIVES

1. Implement a DTW detection system using three different feature extraction methods(STFT, MFCC and LPC). This shows whether the effectiveness of the DTW system is independent from the feature extraction method.
2. Calculate the precision, recall and F1-score for each model.
3. Compare results and determine whether DTW is overall a good detection method and which feature extraction method is best suited.

1.3 SCOPE

What you didn't consider.

1.4 LAYOUT OF REPORT

- * Chapter 2 provides anu background information needed to understand how to solve the given problem as well as any helpful literature of previous solutions to the

problem.

- * Chapter 3 provides an explanation of the overall setup and method used to solve the problem including what metrics were used to meet the stated objectives.
- * Chapter 4 gives a detailed design explanation for all the functional blocks defined in the previous chapter.
- * Chapter 5 provides evidence of results obtained after the system has been designed as well as descriptions of what the results show.
- * Chapter 6 contains all concluding thoughts on the results and what that means for the solutions as well as improvements and recommendations for the project.

My funky thesis (MFT) is as good as MFT can be, as Table 1.1 and Figure ?? shows. Have a look at my explanation in Section 2.1 on page 3, which is in Chapter 2.

Table 1.1: Your table caption

	Centered number [Units]	R-aligned number	Width-set col
Description	100	140	24601

LITERATURE

2.1 RELATED WORK

DATASET

3.1 BACKGROUND

The dataset used contains three 72 hour recordings, recorded in False Bay, near Gordons Bay, between 22 April 2023 and 22 June 2023. The recordings were taken at a depth of 20 to 30 meters at the coordinates, S 34.16813deg, E18.758718deg.

The device used for the recordings was a single omnidirectional hydrophone, designed and built by a Stellenbosch University research group from the Electrical and Electronic Engineering Department.

The recorded data has a sampling rate of 8kHz and a 16-bit resolution. The data was extracted as a 72 hour wav file from an SD card used by the recording device.[1]

3.2 LABELING

Detection tables, containing start and end times of calls within the each file, were provided from a previous research project which used a band-limited energy detector to flag calls. This method unfortunately did not detect all the calls within the audio files and none of the calls were labelled as a specific type of call.

Another detection table was procured. This table contained more detections for the first 72 hour file and contained call labels for all three files. This meant that there were sufficient labels for the first 72 hour recording of the dataset and there were partial labels for the second and third 72 hour recordings within the dataset.[1]

3.3 DATA SPLITTING

For this project the data was split in three sections. Each of the 72 hour audio files was chosen for a different section. Given that only the first of the three 72 hour audio files was

well labeled, it was decided that this would be used for training and testing. The second audio file was used for hyperparameter tuning and the third was used to select templates.

3.3.1 TEMPLATE SELECTION

The audio file which was recorded starting on the 29th was used to select templates. These chosen templates were used for the comparison which determined whether an audio clip contained calls or not.

For the three different call type, single-tones, multi-tones and burst-tonals, the maximum number of templates per call type which could be extracted such that each call type had the same amount of templates was 10. The limiting call type was burst-tonals.

To ensure that the 10 templates chosen were representative of varying amounts of background noise templates were chosen by looking at spectrograms of detected calls throughout the audio file. The visual inspection was also necessary since some of the calls labeled as single-tones looked more like multi-tones and some of the multi-tones could actually be classified as burst-tonals.

3.3.2 HYPERPARAMETER TUNING TEMPLATE SELECTION

The audio file which was recorded starting on the 26th was used to select calibration templates. These templates were used to determine below which threshold a segment of audio could be considered a call. For the calibration templates 10 templates were chosen for each call type. This was again limited by the burst-tonals maximum number of templates. In this file there were however only 9 labeled burst-tonals, but when selecting multi-tones it was found that a call originally labeled as a multi-tone could in fact be seen as a burst-tonal.

Besides these 30 templates, 10 for each call type, 100 background templates were also selected. Within each audio file most of the time is made up of background noise and only some parts actually contain calls. This was simulated by selecting many more background templates than call-type templates. Since there is no table labeling background noise background segments of varying sizes were extracted from the file and 100 templates were manually selected. The reason for the manual process links with the unreliability of the labelled data. The audio file used was only labeled by the band-limited energy detector and thus did not label all the calls within the file. This means that some of the samples extracted as background audio actually contained calls that had never been labeled. Manually selecting each template ensured that all the templates chosen were in fact background noise only.

3.3.3 EVALUATION DATA SELECTION

The entire first audio file which was recorded starting on the 22nd was used for the final evaluation of the whale call detection model. No templates were extracted, instead the entire file was swept to search for calls.

BACKGROUND

4.1 BACKGROUND

Since the late 1990s there have been indications that whales have been struck by shipping vessels.[2] Today this has become a larger concern, as vessel speeds and traffic has substantially increased. In False Bay, South Africa, a colony of Bryde's whales have been detected. Bryde's whales are found to migrate in smaller regions instead of the usual large-scale migrations of other whale species. Due to the increase in shipping traffic as well as the fishing activities in the area, the Bryde's whales in this area are also in danger.[3] Detection and classification systems for whale calls can be used to monitor the migration patterns of the specific whale colony throughout the year. The information gained from such analysis can help determine when vessel traffic should be diverted for the safety of the whale population. These types of systems have been implemented on different whale colonies, but the migration patterns of the False Bay colony has not yet been thoroughly documented.[4] Due to the variations in call types between different colonies of some whale species (killer whales[5], blue whales[6]), models trained for other Bryde's whale colonies would not be completely trustworthy for the detection of calls within the False Bay Bryde's Whale colony as universality of calls cannot be assumed. Therefore, a model trained specifically for the False Bay colony would be beneficial. Dynamic Time Warping (DTW) has been shown to be an effective whale classification method on different whale colonies.[7]

4.2 DYNAMIC TIME WARPING

4.3 FEATURE EXTRACTION METHODS

4.3.1 SHORT TIME FOURIER TRANSFORM

4.3.2 MEL-FREQUENCY CEPSTRAL COEFFICIENTS

4.3.3 LINEAR PREDICTIVE CODING

SYSTEM DESIGN

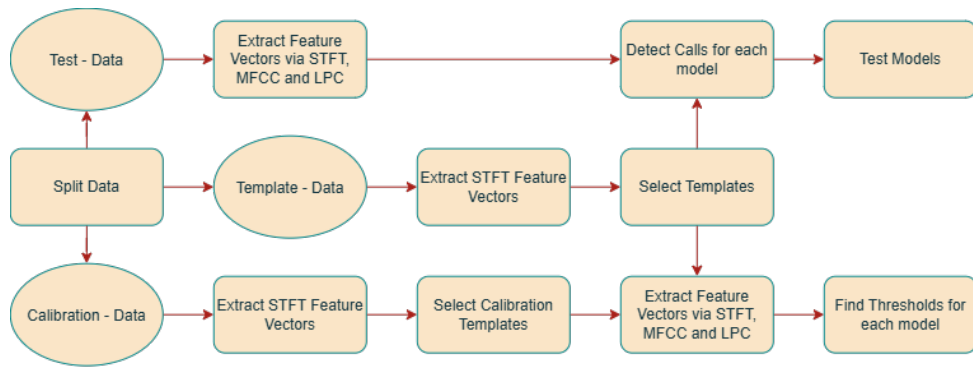


Figure 5.1: Zoomed out image of System Design

5.1 OVERVIEW

5.2 HYPERPARAMETER TUNING

5.3 TRAINING PROCEDURE

5.4 EVALUATION METRICS

5.5 FEATURE EXTRACTION METHODS

RESULTS

6.1 EXPERIMENT 1: STFT FEATURES

6.1.1 5 TEMPLATES

6.1.2 10 TEMPLATES

6.2 EXPERIMENT 2: MFCC FEATURES

6.2.1 5 TEMPLATES

6.2.1.1 NO DERIVATIVES

6.2.1.2 FIRST DERIVATIVE

6.2.1.3 FIRST AND SECOND DERIVATIVES

6.2.2 10 TEMPLATES

6.2.2.1 NO DERIVATIVES

6.2.2.2 FIRST DERIVATIVE

6.2.2.3 FIRST AND SECOND DERIVATIVES

6.3 EXPERIMENT 3: LPC FEATURES

6.3.1 5 TEMPLATES

6.3.2 10 TEMPLATES

CONCLUSION

REFERENCES

- [1] A. Faustmann, “Potential vocalizations of bryde’s whales in false bay, south africa,” 2024, https://scholardata.sun.ac.za/articles/dataset/An_Analysis_of_Potential_Low_Frequency_Vocalizations_of_Southern_African_Bryde_s_Whales_/25108016?file=44351747.
- [2] K. van Waerebeek, A. N. Baker, F. Félix, J. Gedamke, M. Iñiguez, G. P. Sanino, E. Secchi, D. Sutaria, A. van Helden, and Y. Wang, “Vessel collisions with small cetaceans worldwide and with large whales in the southern hemisphere; building a standardized database,” 2006, <https://www.vliz.be/imisdocs/publications/243193.pdf>.
- [3] A. Faustmann, F. W. Shabangu, and J. D. J. J. Versfeld, “Potential vocalizations of bryde’s whales in false bay, south africa,” 2024, <https://repository.up.ac.za/server/api/core/bitstreams/57ba6a31-7aca-4acb-b9b6-60884695c964/content>.
- [4] S. Betz, K. Bohnsack, A. R. Callahan, L. E. Campbell, S. E. Green, and K. M. Labrum, “Reducing the risk of vessel strikes to endangered whales in the santa barbara channel,” 2011, <https://nmssanctuaries.blob.core.windows.net/sanctuaries-prod/media/archive/protect/shipstrike/pdfs/bren.pdf>.
- [5] V. B. Deecke, J. K. B. Ford, and P. Spong, “Call diversity in the north pacific killer whale populations: implications for dialect evolution and population history,” *Animal Behaviour*, 2000, <https://www.sciencedirect.com/science/article/abs/pii/S0003347211005537>.
- [6] M. A. McDonald, S. L. Mesnick, and J. A. Hildebrand, “The geographic variation of blue whale song,” 2003, <https://www.cetus.ucsd.edu/docs/reports/McDonaldIWC-2003.pdf>.
- [7] J. C. Brown, A. Hodgins-Davis, and P. J. O. Miller, “Classification of vocalizations of killer whales using dynamic time warping,” 2006, <https://pubs.aip.org/asa/jasa/article/119/3/EL34/850621>.

APPENDIX A

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